

Module 03: Perception in Robotics

Learning Objectives

1. Define **Perception** within the context of Robotics.
2. Analyze how Perception interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Perception.

Introduction

This module explores **Perception**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Perception is a critical component of the 8-part Active Inference spine, bridging the gap between Agents and Cognition.

Key Concepts

1. Perception as a Markov Blanket Boundary

How does Perception define the boundary between the agent and the environment?

2. Generative Models of Perception

What parameters involved in Perception must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Perception drive the perception-action loop?

Applications

In Robotics, we see Perception manifest in: * **Specific Example 1:** A self-driving car's perception stack fuses LIDAR point clouds, stereo camera images, and radar returns through a deep-learning-based BEV (bird's-eye view) transformer that outputs a probabilistic occupancy grid and tracked object list; this multi-sensor fusion implements generative model inversion where each sensor modality provides a distinct likelihood function over the scene state, and the network's learned feature representations encode the observation model that maps 3D world states to expected sensor measurements across all modalities simultaneously. * **Specific Example 2:** An autonomous underwater vehicle (AUV) performing pipeline inspection uses forward-looking sonar as its primary perceptual modality in turbid water where cameras fail; the AUV's generative model predicts expected sonar returns given hypothesized pipe positions and orientations, and perception consists of inverting this acoustic model to infer pipe geometry -- when the sonar image deviates from predictions (e.g., a damaged section with unexpected geometry), the prediction error signals an anomaly that triggers closer inspection, demonstrating active perception driven by surprise.

Conclusion

Understanding Perception allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.